R16

Code No: 138CV

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July/August - 2022
INFORMATION THEORY AND CODING
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Explain briefly about Average and Mutual Information and Entropy.
b) Write a brief note on Types of Errors. [9+6]

2.a) Explain in detail about Error Control Strategies.
b) Discuss about the Mathematical model of Information. [8+7]

3. Discuss about Linear Block Codes and write a detailed note on Applications of Block codes for Error control in data storage system. [15]

4.a) Write a short note on Standard array and Syndrome Decoding.
b) Explain Probability of an undetected error for Linear Codes over a BSC. [7+8]

5.a) Write a detailed note on Error-trapping decoding for cyclic codes.
b) Explain about Majority logic decoding for cyclic codes. [8+7]

6.a) Explain briefly about Encoding.
b) Write a detailed note on Syndrome Computation and Error Detection. [5+10]

7. What are Convolutional Codes? Explain in detail the application of Viterbi Decoding and Sequential Decoding. [15]

8.a) Write a brief note on Syndrome computation and iterative algorithms.
b) Explain about Error locations polynomials for single and double error correction. [7+8]

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